Patient Flow Simulator — Decision Report

Type 1 Emergency Department

Executive summary

- Modelled demand: 325 attendances/day (1,952 patients over the simulated period).
- 4-hour standard (UEC Plan 2025/26 target >=78%): 79% [MEETS] (95% CI 70-80% across 24 Monte-Carlo runs) - 417 breaches.
- 12-hour waits (UEC Plan target <10%): 1.2% [MEETS] - 24 patients waited over 12 hours.
- Mean time in department: 2.8 h (95% CI 2.8-3.4 h); 95th percentile 8.2 h.
- Admitted: 16% of attendances.

Bottleneck diagnosis

Biggest delay: SDEC — patients wait 111 min on average (681 patient-hours lost over the run).

Recommendation: sdec doctor is busy (75% utilisation) and likely drives this queue; trial extra capacity in Scenario comparison.

What-if scenarios (ranked)

Scenario	4-hr perf.	vs baseline	Mean LOS
Baseline	75%	—	3.0 h
+3 sdec doctor	85%	**+10 pts**	2.5 h
+8 inpatient bed	75%	**+0 pts**	3.0 h
Winter +15% demand	70%	**-5 pts**	3.5 h

For demonstration — synthetic data seeded from the beta config.json. Validate against local data before operational use.

Patient Flow Simulator

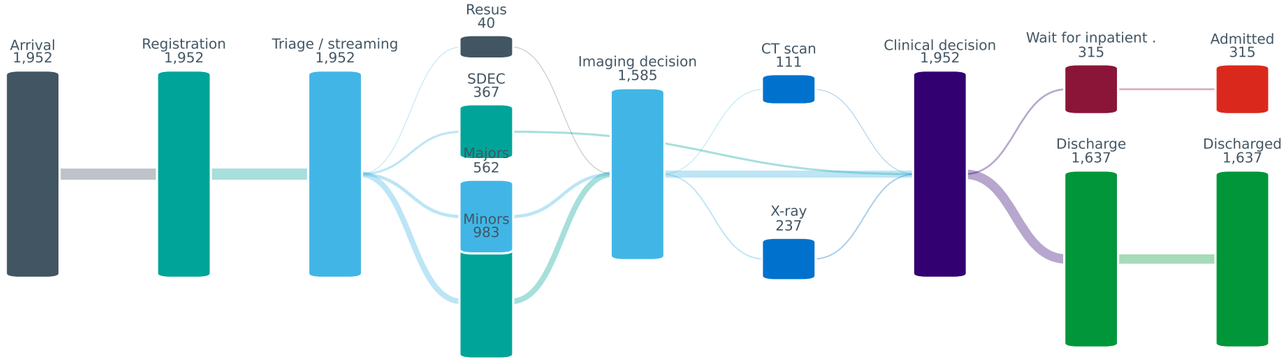
Discrete-event simulation for emergency & urgent care planning

Type 1 Emergency Department

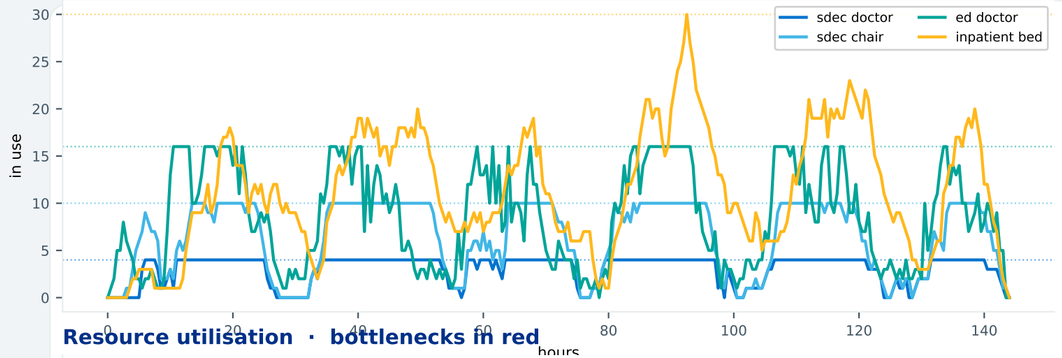
325 attendances/day · 6-day run · 24 Monte-Carlo replications

4-HOUR PERFORMANCE	MEAN TIME IN DEPT	95TH PERCENTILE LOS	ADMITTED	DAILY ATTENDANCES
79%	**2.8h**	**8.2h**	**16%**	**325**
95% CI 70-80%	95% CI 2.8-3.4h	longest typical wait	7.2h mean LOS	1,952 simulated

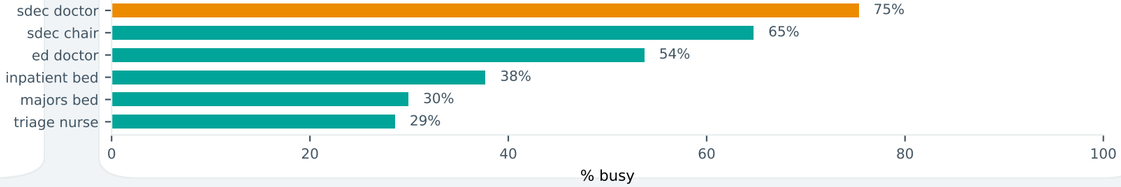
Patient flow · arrival to disposition



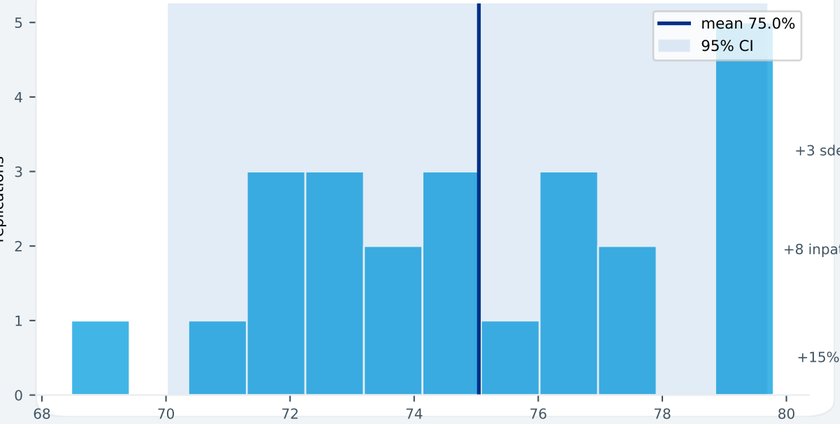
Resource occupancy vs capacity (dotted)



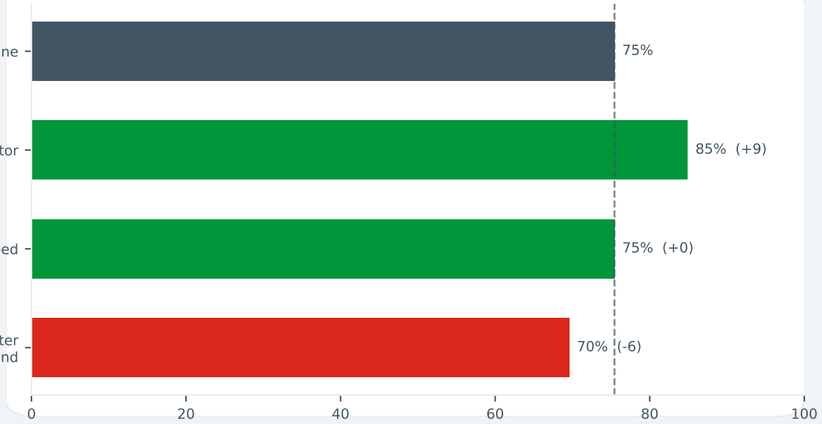
Resource utilisation · bottlenecks in red



Monte-Carlo: 4-hour performance variability



Scenario comparison · 4-hour performance



Where patients wait · mean queue time

STAGE	WAIT	SEEN
SDEC	**111m**	367
CT scan	**8m**	111
Resus	**4m**	40
Minors	**3m**	983
Majors	**3m**	562
X-ray	**1m**	237

For demonstration — synthetic data